

Matthew Yu

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SKILLS

Languages	Python, Java, C , C++, TypeScript/JavaScript, SQL, Lua, Racket, Ruby (Ruby on Rails), OpensCAD
Web	React, React Native, Next.js, Node.js, Express, REST APIs, WebSockets, Flask, Spring Boot, Auth, Web Assembly
Data/ML	MongoDB, MySQL, Web Scraping (Selenium, BeautifulSoup), Tensorflow (Keras), Pytorch, OpenCV, Parser
Tools	Git, SSH, Linux/Unix, Bash, Postman, Vim, LaTeX, JIRA, Confluence, Docker, Kubernetes

EXPERIENCE

Reality Labs Design Team – <i>Software Engineer (Humanoid Team)</i>	<i>Sept 2025 – Present</i>
Technologies: JavaScript, C#, Express, WebSockets, Unity	Waterloo, ON
• Developed backend services for a Unity VR application using Express and WebSockets , achieving sub-50 ms end-to-end interaction latency from VR Input to App	
Project: Empower – <i>Software Engineer (Backend)</i>	<i>Aug 2024 – Aug 2025</i>
Technologies: Python, Flask, BeautifulSoup	Remote
• Built reliable Flask backend for app that collects student opportunities from external web sources using Beautiful Soup .	
Mu Systems Corporation – <i>Software Engineer (Backend)</i>	<i>Jul 2024 – Aug 2024</i>
Technologies: Java, Spring Boot, Python, BeautifulSoup, Zentao	Remote
• Developed backend features for internal tools at an e-commerce startup , including route optimization for delivery drivers, item search using Elasticsearch , and automated competitor price collection via web scraping .	

AWARDS & QUALIFICATIONS

Competitive Programming — Codeforces (Top 10%), DMOJ (Top 700), USACO Silver	2021 – Present
10x Hackathon Participant (2x Winner)	2024 - Present
3rd Place — Forester Pitch Challenge at UofT Hacks	2026
1st Place — PTC Hacks	2024
SHAD Program — Best Design Award for Website Creation	2024

PROJECTS

Bricked — Constraint-Aware LEGO Design Platform	 
<i>Next.js, React, Three.js, OpenCV, OpenSCAD, FastAPI, LangGraph, WebAssembly, Webworker, Manifold</i>	
• Digitized real LEGO inventory via computer vision for accurate 3D reconstruction	
• Generated LEGO builds using a multi-agent AI pipeline (LangGraph) for quality and based on scanned bricks	
• Built & Optimized an OpenSCAD compiler (WASM + Manifold), reducing render times	
• Delivered interactive, step-by-step 3D build instructions with per-brick customization in DIY Solidworks-like platform	
Focus Friend — EEG-Based Focus Detection System	 
<i>React Native, TypeScript, C++, Hardware (Arduino / Operational Amplifiers)</i>	
• Constructed a mobile app that streams EEG-derived signals and visualizes focus vs. fatigue in real time	
• Implemented a data pipeline for preprocessing noisy sensor input and running on-device/in-app inference	
• Integrated hardware acquisition with a React Native UI and Bluetooth , to deliver feedback loops for users	
Philos — 3D Personable Friend	 
<i>Next.js, Three.js, OpenCV, Azure, TensorFlow, Tailwind</i>	
• Built a web app combining a 3D interface with computer vision to detect emotion for interactive user experiences	
• Implemented secure session handling using made-from-stratch cookie-based authentication and protected API routes	

EDUCATION

University of Waterloo	Present
Bachelor of Computer Science (Honours)	Waterloo, ON
GPA: 4.0 • Clubs: CS, DS • Advanced Math 145 & CS 146 • VP of Operations, Residence Council	
Dr. G. Williams Secondary School	
IB Diploma, OSSD, French Merit	Aurora, ON
GPA: 97.7% • President: Math Club • Executive: Game-Dev & Peer Tutors • Clubs: Code, Debate	
• 2x Qualifier for OFSSA Cross Country & Captain • Certified National Lifeguard & Swim Instructor	